

TRANSFORMER

Faraday's Law:-

Faraday 1st law :-

When the magnetic flux linked in a coil or conductor changes an emf is induced in that coil or conductor.

Faraday 2nd law :- The magnitude of induced emf in coil or conductor is proportional to the rate of change of the magnetic flux linking that coil or conductor.

$$e \propto \frac{d\phi}{dt}$$

 ϕ = magnetic flux $\frac{d}{dt}$ = rate of changeand e = emf

→ Where ϕ is the magnetic flux linkage of the CKT or coil. If coil has 'N' turns, flux linkage with the coil is 'N ϕ ' and magnitude of EMF is

$$e \propto \frac{d(N\phi)}{dt} \Rightarrow e = K \frac{d(N\phi)}{dt}$$

→ Where K is constant of proportionality in ϕ is

→ The SI unit of K is 1

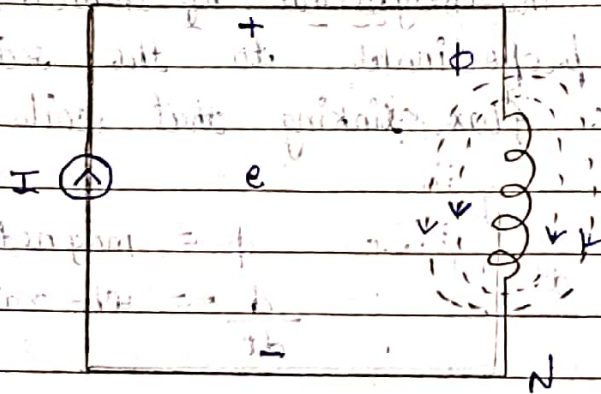
$$e = \frac{d(N\phi)}{dt}$$

★ ★ → The direcⁿ of induced ~~emf~~ ^{emf} is given by Lenz's law.

→ Lenz's law states that the direcⁿ of induced emf is always oppose the cause that produce it.

$$\star \quad \mathcal{E} = -\frac{d(N\Phi)}{dt}$$

Self inductance :-



→ Consider a single inductor, a coil with N turns. When current ' I ' flows through the coil the magnetic flux Φ is produced around it. According to Faraday's law -

$$\mathcal{E} = -N \frac{d\Phi}{dt} \quad \text{but the flux } \Phi \text{ is}$$

produced by current ' I '. So that any change in Φ is due to change in the current.

$$\mathcal{E} = -N \frac{d\Phi}{dt} \frac{dI}{dt}$$

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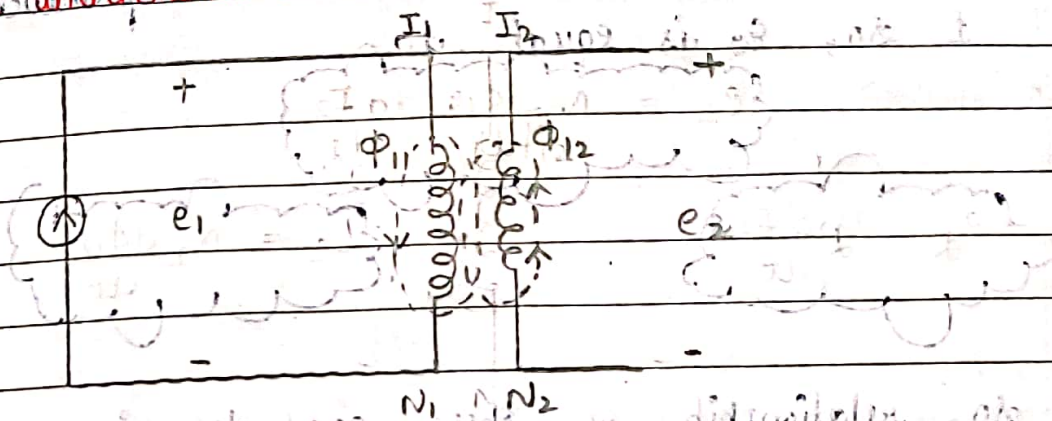
$$e = L \frac{dI}{dt}$$

$$L = N \frac{d\phi}{dI}$$

This equation gives voltage current relationship for the inductor and Here L is the inductance of the inductor. This inductance is known as self inductance.

$$\star L = N \frac{d\phi}{dI}$$

Mutual inductance :-



In mutual inductance two coils with self inductance L_1 and L_2 that are in close proximity to each other. Coil 1 has N_1 turns and coil 2 has N_2 turns. The second inductor carries no current the total flux of the circuit is represented as

$$\phi = \phi_{11} + \phi_{12}$$

* The emf of coil No. 1 is $\rightarrow$

$$e_1 = N_1 \frac{d\phi_{11}}{dt}$$

* The emf of coil No. 2 is $\rightarrow$

$$e_2 = N_2 \frac{d\phi_{12}}{dt}$$

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The flux is caused by the current

$$e_1 = N_1 \frac{d\phi_{11}}{dt} \cdot \frac{dI_1}{dt}$$

$$e_1 = L_1 \frac{dI_1}{dt}$$

$$L_1 = N_1 \frac{d\phi_{11}}{dt}$$

In coil No. 2 the flux is produced by coil No. 1 so, e_2 is equal to

$$e_2 = N_2 \frac{d\phi_{12}}{dt} \cdot \frac{dI_1}{dt}$$

$$e_2 = L_2 \frac{dI_2}{dt}$$

$$L_2 = N_2 \frac{d\phi_{12}}{dt}$$

So, relationship of these conductor is -

$$\begin{aligned} M_{11} &= M_{12} = M \\ e_{11} &= e_{12} = e \\ \phi_{11} &= \phi_{12} = \phi \end{aligned}$$

Coefficient of coupling is:-

$$M = k \sqrt{L_1 L_2}$$



TRANSFORMER :-

Transformer is an electrostatic device which is used to transfer electric energy (voltage and current) from one CKT to another CKT by mutual induction of two electric CKT, without change in frequency which is working under the principle of electro magnetic induction.

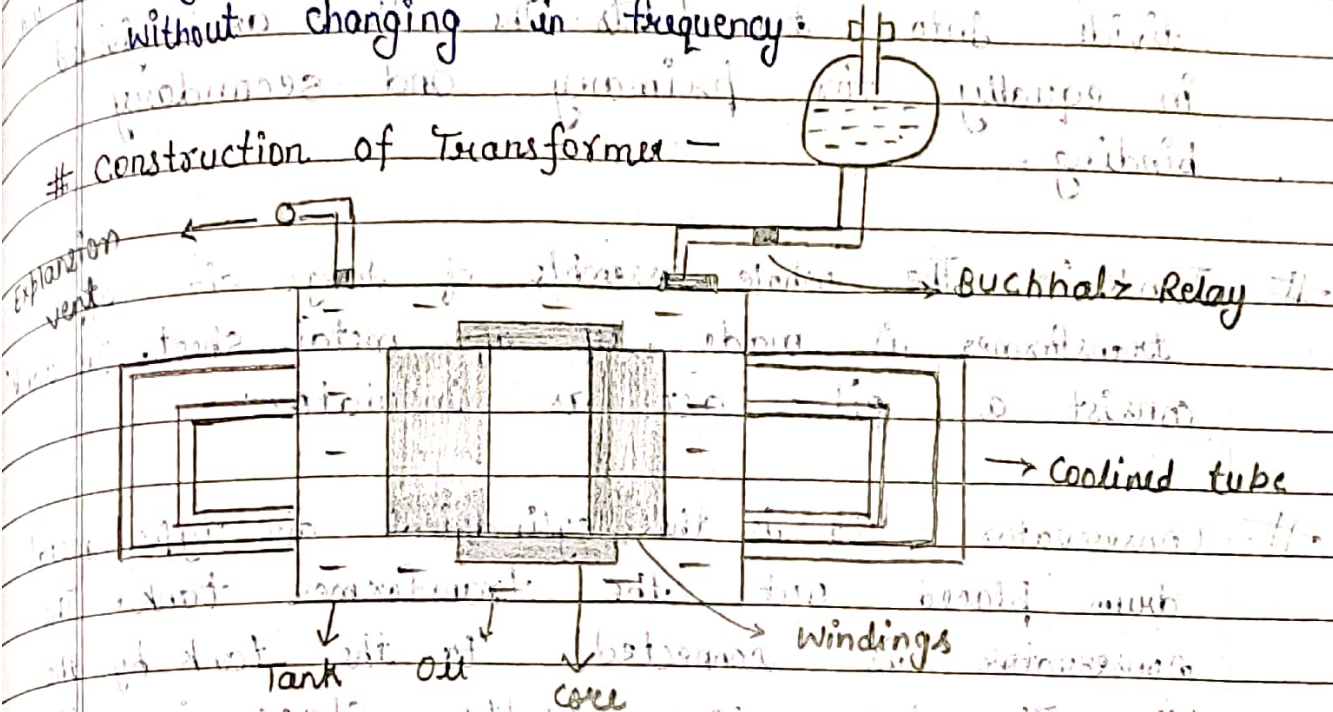
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Transformer is the static device which transfer to voltage or current from one level to another level without changing in frequency.

Construction of Transformer -



The most important part of transformer is binding and cores.

Core :- The core is made up of steel lamination so that to minimise the eddy current. The lamination is about 0.3 mm thick and insulating from each other. According to core there are two types of transformer -

(i) Core type

(ii) Shell type

Core type transformer: In this type coil is bound around the two limbs of rectangular magnetic core. Each limb carry one half of primary binding and one half of secondary boundary known as core type transformer.

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Shell type transformer:- It has three limbs and both the windings are placed on the central limb the entire flux passes through the central limb and divide into two parts. The flux are divided in equally for primary and secondary winding.

Tank :- The whole assemble of large size of transformer is made up of metal sheet. In tank consist a oil act as insulator.

Conservator :- It is the cylindrical air type metal drum placed out the transformer tank. The conservator is connected to the tank by the pipe. It has some empty space is available above the oil.

Breather :- When oil temperature increases it expands. In air it will be out of the conservator.

Buchholz relay :- It is a protective device mounted on the pipe connecting tank and conservator is called buchholz relay.

Explosion vent :- It consist of a glass diaphragm. When the transformer becomes faulty the cooling oil will get decomposed known as explosion vent.

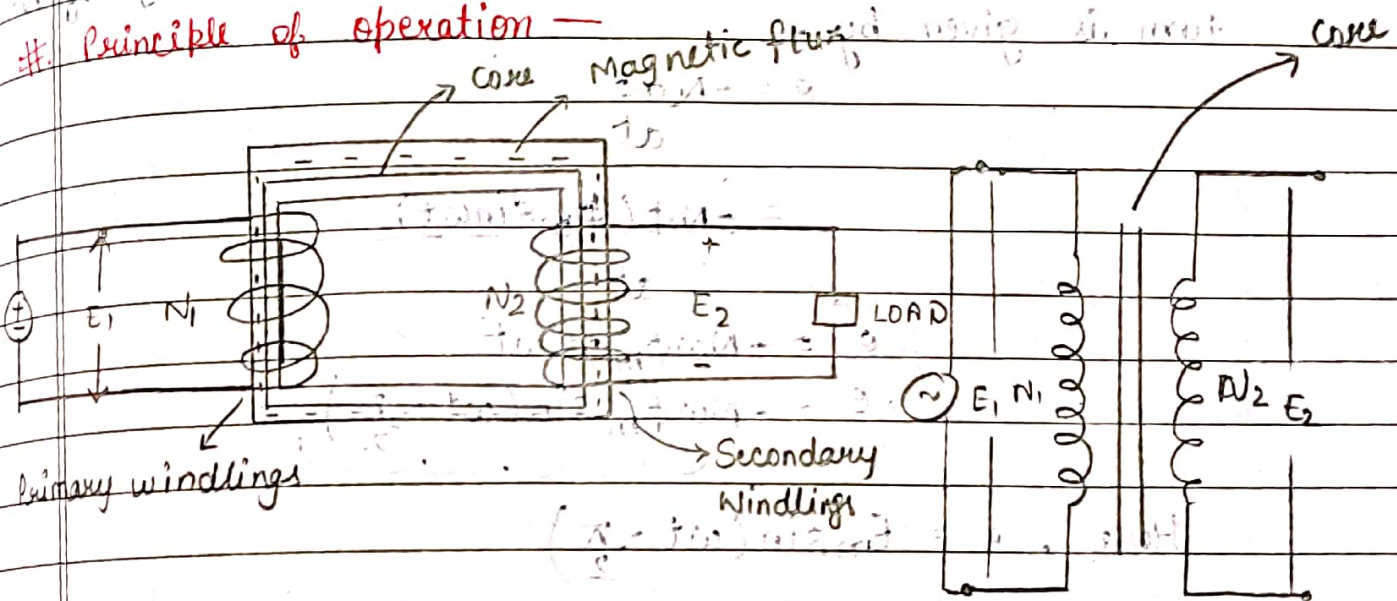
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Ideal transformer condition :-

- (i) The permeability (μ) of the core is infinite.
- (ii) There is no losses due to resistance, hysteresis, eddy current.
- (iii) The resistance of winding is zero hence $I^2 R$ losses will be zero.
- (iv) There is no leakage flux.

Principle of operation -



→ When one coil is connected to an AC supply of definite voltage and frequency then alternating flux is set up in the laminated core. The flux is link with the other coil and produce mutually induce emf in that coil.

According to Faraday's law's

$$e = -M \frac{dI}{dt}$$

When N_2 is greater than N_1 the transformer called as step-up transformer.

When N_1 is greater than N_2 the transfor is called as step-down transformer.

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#. EMF equation of transformer —

It can be present as

$$\phi = \phi_m \sin \omega t$$

$$\phi = \phi_m \sin 2\pi f t$$

Here ϕ_m is the peak value of the flux.
 f is the frequency of sinusoidal variation of the flux. As per the ~~elect~~ law of electromagnetic induction. The induced emf in a winding of N -turn is given by $\rightarrow$

$$e = -N \frac{d\phi}{dt}$$

$$= -N \frac{d(\phi_m \sin \omega t)}{dt}$$

$$\left. \begin{aligned} e &= -N \omega \phi_m \cos \omega t \\ e &= -N \omega \phi_m \sin \left(\omega t - \frac{\pi}{2} \right) \end{aligned} \right\}$$

$$\text{Here, } e = E_m \sin \left(\omega t - \frac{\pi}{2} \right)$$

$$E_m = -N \omega \phi_m$$

E_m = maximum value of induced emf

The rms value of induced emf is given by —

$$E = \frac{E_m}{\sqrt{2}} = \frac{\omega N \phi_m}{\sqrt{2}} = \frac{2\pi f N \phi_m}{\sqrt{2}} = 4.44 f N \phi_m$$

This eqⁿ is known as emf equation of transformer

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→ RMS value of emf induced primary winding.

$$E_1 = 4.44 f \phi_m \times N_1$$

→ RMS value of emf induced in secondary winding.

$$E_2 = 4.44 f \phi_m \times N_2$$

Voltage transformation Ratio of transformer -

$$K = \frac{V_2}{V_1} = \frac{N_2}{N_1} = \frac{E_2}{E_1}$$

→ When $K > 1$, so $N_2 > N_1$ the transformer is known as step-up transformer.

In this case voltage 2 volts secondary side is higher.

→ When $K < 1$ so $N_2 < N_1$ the transformer is known as step-down. In this case voltage 2V towards primary side is higher.

Volt - Ampere Ratio of ideal transformer -

$$I_1 N_1 = I_2 N_2$$

$$\star \star \star \quad \frac{I_2}{I_1} = \frac{N_1}{N_2} = \frac{1}{K}$$

$$\star \star \star \quad \frac{E_2}{E_1} = \frac{I_2}{I_1} = \frac{N_1}{N_2} = \frac{1}{K}$$

Transformer losses : — Transformer is the static device not a rotating device. Therefore frequency friction and winding losses are not present. The power losses of

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Transformer is two types like as copper loss and Iron loss.

- (i) Copper loss :- When ever current flows through any conductor there is power loss. The magnitude of this loss is proportional to the square of the current flowing through.

$$P_C = I_1^2 R_1 + I_2^2 R_2$$

- (ii) Iron loss :- These losses take place in the iron part of in the core area of the transformer. The Iron loss is the two types —

- (i) Hysteresis loss (ii) Eddy current loss

Hysteresis loss :- This loss is due to the reversal of the magnetisation of the transformer core. This loss depends on volume and grade of iron.

$$W_H = \eta B_m^{1.6} f V$$

Watt.

Eddy current loss :- Due to alternating flux in the core eddy current will flow in the coil. Power is required to maintain these eddy current. This power is dissipated in the form of heat and is known as eddy current loss.

$$W_e \propto B_{max}^2 f^2 t^2$$

t is the thickness of the core.

Efficiency of transformer :-

$$\text{efficiency } \eta = \frac{\text{Output Power}}{\text{Input Power}} = \frac{\text{Output Power}}{\text{Output Power} + \text{Losses}}$$

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Ques. An ideal transformer has a turns ratio of 8:1 and the primary current is 3A. When it is supplied at 240V. Determine the 2nd V and I?

Solⁿ → Given — find —

$$N_1 = 8$$

$$V_2 = ?$$

$$N_2 = 1$$

$$I_2 = ?$$

$$I_1 = 3$$

$$V_1 = 240$$

$$\frac{I_1}{I_2} = \frac{N_2}{N_1}$$

$$= \frac{3}{I_2} = \frac{1}{8} \Rightarrow I_2 = 24A$$

$$I_2 = 24A$$

$$\frac{V_2}{V_1} = \frac{N_2}{N_1}$$

$$\frac{V_2}{240} = \frac{1}{8} \Rightarrow 8V_2 = 240$$

$$V_2 = 30V$$

Ques. An ideal transformer connected to a 240 volt has turns ratio is equals to 9:2 and the primary current is 4A. When it is supplied at 270V. 2nd V & I?

$$\Rightarrow \frac{I_1}{I_2} = \frac{N_2}{N_1} \Rightarrow \frac{4}{I_2} = \frac{2}{9} \Rightarrow 2I_2 = 36$$

$$I_2 = 18A$$

$$\frac{V_2}{V_1} = \frac{N_2}{N_1} \Rightarrow \frac{V_2}{270} = \frac{2}{9} \Rightarrow 9V_2 = 540$$

$$V_2 = 60V$$

Ques. A transformer has 540 primary turns and 3000 2nd turn. If 1st V is 240V & Determine the 2nd V assuming Ideal condition

$$\frac{3000}{540} = \frac{V_2}{240} \Rightarrow V_2 = \frac{3000 \times 240}{540} = 1333.33V$$

$$\frac{72000}{54} = V_2 \Rightarrow V_2 = 1333.33V$$



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Q. Two identical 750 turns coil A & B are in // plane. A current is changing at the rate of 1500 A/s in coil A. An induced emf of 11.25 V in coil B. Find the mutual inductance of this arrangement. If the self inductance of each coil is 15 mH. Find the flux produced in coil A per ampere and the % of this flux which links the turn of coil B.

Solⁿ ⇒ Given —

$$N_1 = N_2 = 750$$

$$\frac{dI}{dt} = 1500 \text{ A/s}$$

$$e_2 = 11.25 \text{ V}$$

$$L = 15 \text{ mH}$$

Find —

$$M = ?$$

$$\phi_1 = ?$$

$$\Rightarrow \text{OVS} = e_2 = M \frac{dI}{dt}$$

$$M = \frac{e_2}{\frac{dI}{dt}}$$

$$M = \frac{11.25}{1500}$$

$$M = 7.5 \times 10^{-3}$$

$$\text{OVS} = e_1 = N_1 \frac{d\phi_1}{dt}$$

$$\phi_1 = \frac{e_1}{N_1 \frac{dI}{dt}}$$

$$\phi_1 = \frac{11.25}{750 \times 1500}$$

$$= \frac{1}{5} \times 10^{-4} = 0.2 \times 10^{-4}$$

$$\frac{d\phi}{dt} = 0.2 \times 10^{-4} = 2 \times 10^{-5}$$

$$\frac{\phi}{I} = 2 \times 10^{-5} \text{ Wb/A}$$

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$$M = K \sqrt{L_1 L_2}$$

$$\frac{M}{\sqrt{L_1 L_2}} = K$$

$$K = \frac{7.5 \times 10^{-3}}{\sqrt{225 \times 10^{-6}}}$$

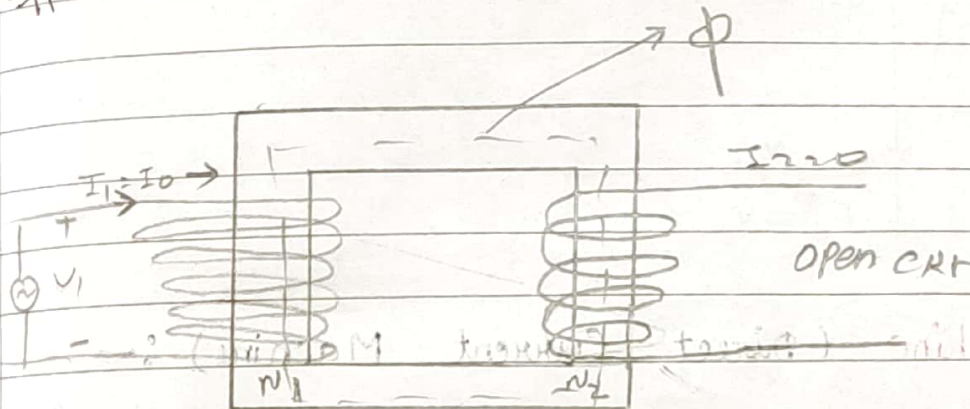
$$= \frac{75 \times 10^{-3}}{225 \times 10^{-5}} = \frac{3 \times 10^2}{5}$$

$$K = 0.5$$

$$K = 50\%$$

$$0.5 \times 10^2$$

Transformer At no load :-



When transformer is on no load it is connected to supply with the secondary open CKT, current I_0 will set up a flux in the core. The no load condition of the transformer has measure iron loss (Hysteresis and eddy current). The current I_0 will resolve the resultant component of hysteresis and eddy current in the transformer.

Hysteresis current is denoted as $I_H = I_0 \cos \phi_0$

& eddy current is denoted as $I_{eddy} = I_0 \sin \phi_0$

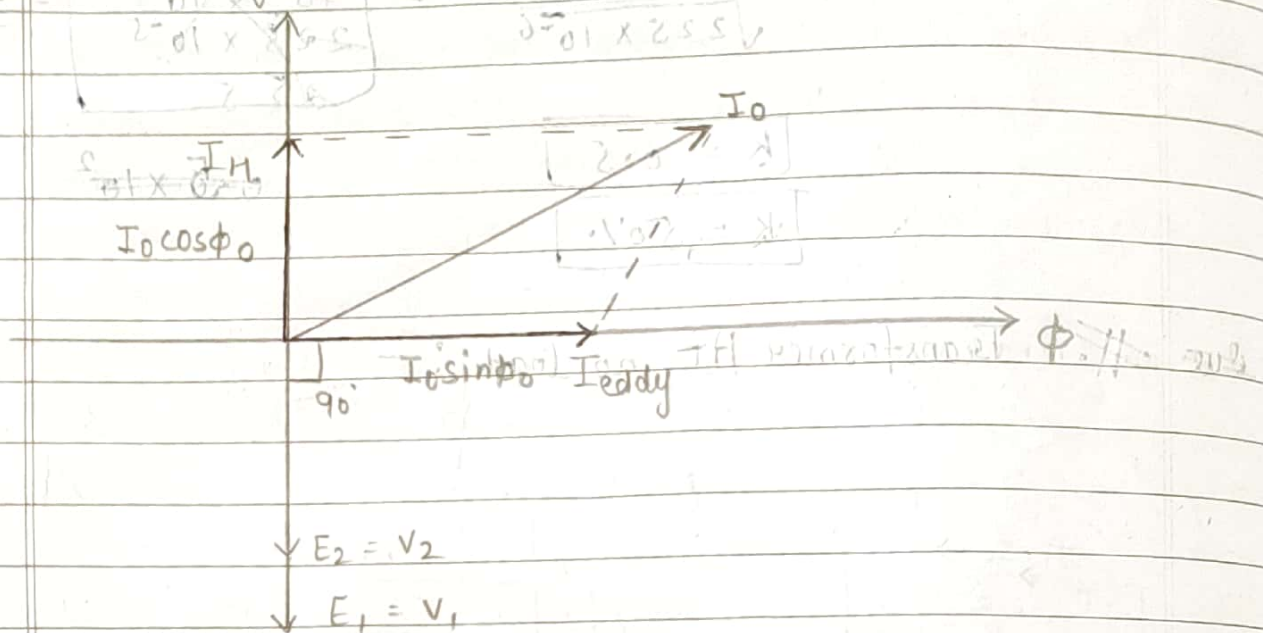
Since I_0 is the vector sum of I_H and I_{eddy} ,

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$$I_0 = \sqrt{I_H^2 + I_{\text{eddy}}^2}$$

→ So, represented as phasor diagram transformer at no load.



DC Machine (Direct Current Machine) :-

A DC machine

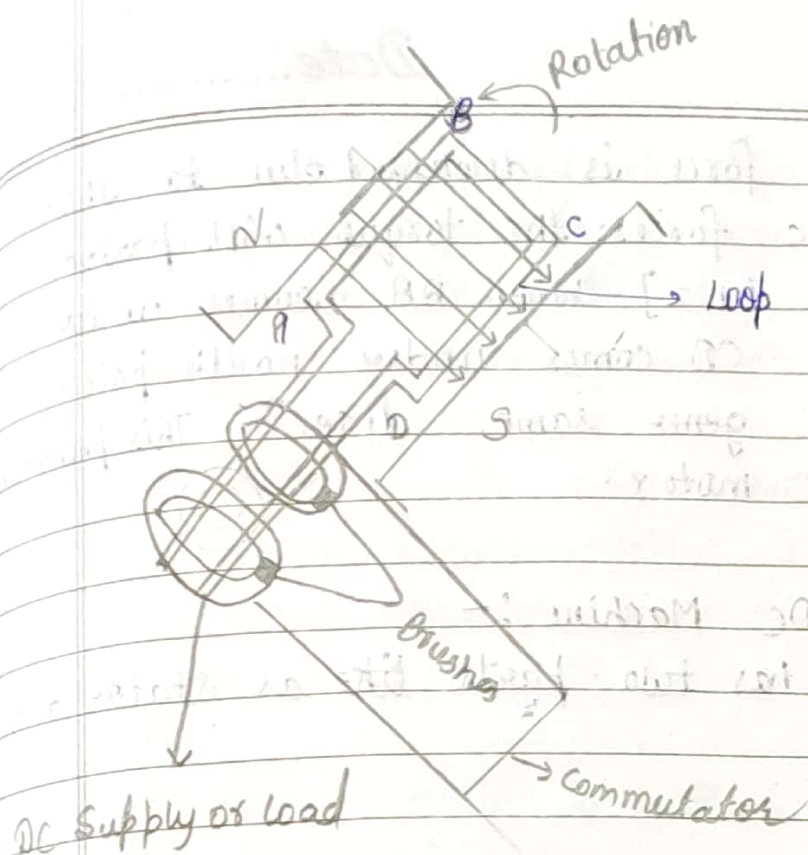
is a electromechanical energy conversion device. An electrical machine which converts mechanical energy into a electrical energy of a DC nature is called DC Generator. An electrical machine which convert an electrical DC power into mechanical energy is called as DC motor.



Working principle of DC Machine :-

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The shown in figure can work as DC generator and DC Motor. A single loop made of two conductors AB and CD is placed in magnetic field. Loop can rotate along the axis.

Generating Mode → When the loop is rotated by supplying mechanical energy. It cuts the line of magnetic field and emf is induced in loop.

Induced emf is alternating in nature and this alternating current is uni direction. This phenomena is represent as DC generator.

Rotating Mode :- When electric current (supplying electric energy) passes through a loop. It feels a force. Direcⁿ of this force is governed by Fleming left-hand rule. In conductor CD the force is upward.

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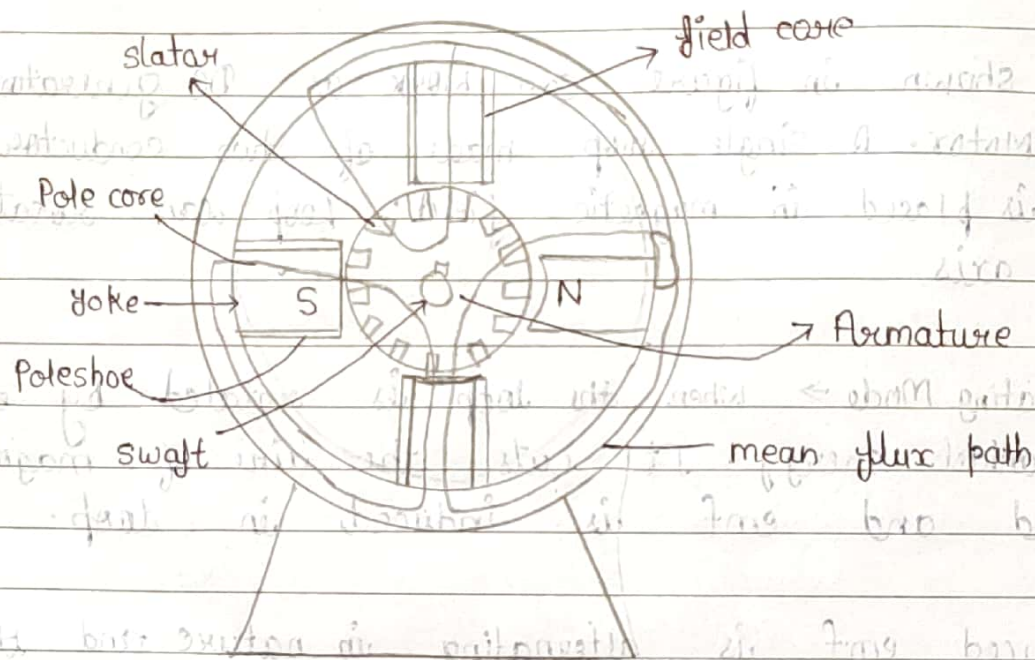
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and conductor AD force is downward due to these couple of magnetic forces. the torque will produce. ($F \times L$) [given by lenz]. When AB comes under South pole and CD comes under north pole. The torque will give same direcⁿ. This phenomena is known as DC motor.

* Construction of DC Machine :-

DC Machine has two parts like as stator and rotor.

1. Stator :-



Stator :- It is a stationary part and is designed mainly for producing a magnetic flux.

Yoke :- It is the outermost cylindrical part which serve two purpose.

It provide the supporting of the frame and second is provide the path of magnetic flux.

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Poles :- Pole core carries a field winding which is necessary to produce magnetic flux.

It is spread out the magnetic flux produce through air gap to armature core.

Field coil :- The coil wound around the pole cores are called field coil. It carries the current due to which the pole coil behaves as an electro-magnetic.

2 Rotar :- Rotar is the rotating part it is inner cylindrical part having armature and commutator brush arrangement.

(easy path) Armature :- It is cylindrical in shape and provides low reluctance for magnetic part.

Commutator :- It is a most important part of DC machine. It serves the following purpose —

1. It connect the rotating armature conductors to the stationary external ckt through brushes.
2. It converts AC induced in the armature conductors into uni directional current in the external load ckt.

3 Brushes :- It is use to induce emf in the armature are connected to the commutator segment.

4 Bearings :- The ball bearing is used to both ends of DC Machine.

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Swag: - The rotating part such as armature core, commutator, cooling fan are used as swag. It is transfer to the mechanical power form to the machine.

Field: - The field winding is placed on the pole pieces of the machine. It carries the field current and produces the magnetic field.

Rotor: - Rotor is the rotating part of the machine. It carries the armature winding and commutator.

Armature: - It is cylindrical in shape and carries the armature winding. It is the main part of the machine.

Commutator: - It is a most important part of the machine. It carries the armature winding and provides the electrical connection between the armature and the external circuit.

Brushes: - They are used to make contact with the commutator segments and carry the current to the external circuit.

Field winding: - It is used to produce the magnetic field in the machine. It is connected to the DC supply.

Brushing: - The process of removing the carbon dust from the commutator surface is called brushing.

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